Ahmed Hany AlHady

SOFTWARE ENGINEER | FULL-STACK DEVELOPMENT | CLOUD AND AI

Alexandria, Egypt | +20 127 651 9991 | ahmedhanyx115@gmail.com | GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Computer Engineering graduate with dual bachelor's degrees from AASTMT and the University of Northampton. Builds end-to-end software using JavaScript, Python, C#, MERN, FastAPI, REST APIs, relational and NoSQL databases, and cloud-native delivery workflows. Experience spans backend systems, full-stack web development, computer vision, IoT, and cybersecurity, with a record of leading technical work and collaborating across engineering teams.

TECHNICAL SKILLS

Programming: C, C++, Python, C#, Java, JavaScript (ES6+), SQL
Backend and Web: Node.js, Express.js, React.js, HTML, CSS, FastAPI, .NET, REST APIs
Cloud and DevOps: AWS, Docker, Kubernetes, CI/CD pipelines, Agile, Scrum
Testing: Jest, PyTest, Cypress
AI, Data and IoT: Computer vision, deep learning, ResNet, CNNs, PyTorch, Apache Spark, embedded systems, IoT architecture
Cybersecurity: Penetration testing, vulnerability assessment, SQL injection, XSS, CSRF, network security, threat analysis

PROFESSIONAL EXPERIENCE

Full Stack Developer | Andalusia | Alexandria, Egypt | Jul 30, 2026 - Present

- Build a full-stack academy platform that connects a responsive React 18 and TypeScript experience with ASP.NET Core (.NET 10) APIs, EF Core, and SQL Server.
- Deliver reusable public-site features for course and program discovery, bilingual content, and responsive RTL/LTR support.
- Strengthen delivery with versioned REST APIs, Swagger/OpenAPI, Identity and database foundations, Docker-based development, EF Core migrations, and automated tests.

Backend Software Engineer | FlyRank | Remote | Jul 2026 - Present

- Design, build, and maintain scalable REST APIs in Node.js, Python, and JavaScript for high-traffic web applications.
- Architect and deploy AWS-hosted microservices that improve system reliability and fault tolerance.
- Partner with frontend engineers and product managers to define requirements and deliver cohesive user experiences.

IT Intern | Amreya Petroleum Refining Company | Alexandria, Egypt | Aug 2025

- Resolved system and network issues to help maintain 99.999% operational uptime for critical refining processes.
- Streamlined software updates and data-management tasks, improving reliability and security across departments.
- Worked with cross-functional teams to improve IT workflows, efficiency, and response time in a high-demand industrial environment.

AI and Cybersecurity Intern | Universitat Autonoma de Barcelona | Barcelona, Spain | Jul 2025

- Completed an intensive program in generative AI, large language models, natural language processing, and knowledge representation.
- Applied threat detection, data protection, and secure AI deployment practices through hands-on workshops.
- Collaborated with international experts at INCIBE on AI and critical-infrastructure security.

Computer Engineering Trainee | UCLan University Engineering Summer School | Preston, UK | Sep 2024

- Built a water-rescue drone using a MAX30205 thermal sensor, Arduino, and a data-acquisition system.
- Earned 2nd place for Best Intelligent Machine Project and strengthened skills in embedded systems and sensor integration.

EDUCATION

Bachelor's Degree in Computer Engineering | University of Northampton | Northamptonshire, UK | Jun 2026

Dual-degree program with the Arab Academy for Science, Technology and Maritime Transport.

Bachelor's Degree in Computer Engineering | Arab Academy for Science, Technology and Maritime Transport | Alexandria, Egypt | Jun 2026

LANGUAGES

Arabic - Native | English - C1 | German - A2

SELECTED PROJECTS

Andalusia Academy | Bilingual Education and Learning Platform

Role: Full-Stack Developer **| Technologies:** React 18, TypeScript, Vite, ASP.NET Core (.NET 10), EF Core, SQL Server, REST APIs, Docker

- Develop an end-to-end academy platform connecting public course and program discovery with foundations for authentication, learner access, and administrative operations.
- Combine a reusable React and TypeScript frontend with versioned ASP.NET Core APIs, SQL Server persistence, automated tests, and bilingual RTL/LTR support.

Alzemora - Smart Glasses and Mobile Ecosystem for Alzheimer's and Dementia Patients | Graduation Project

Role: Software Engineering Team Lead, Mobile and AI Developer **| Technologies:** React Native, Python, MERN, REST APIs, IoT, computer vision, speech recognition, Figma | GitHub

- Led development of an AI-powered wearable and companion application supporting patients and caregivers.
- Built a React Native caregiver dashboard for real-time monitoring, location tracking, and emergency alerts.
- Integrated face and voice recognition, IoT fall detection, and AES-256 encrypted communication between the smart glasses and cloud.

EuroSAT Land Use Classification | AI and Computer Vision

Role: AI Engineer **| Technologies:** Python, PyTorch/TensorFlow, ResNet-50, CNNs

- Developed a deep-learning model to classify EuroSAT satellite imagery across multiple land-use categories.
- Fine-tuned a pre-trained ResNet-50 with transfer learning and data augmentation, then evaluated performance using confusion matrices and precision-recall metrics.

Large-Scale Data Analytics Pipeline | Big Data

Role: Big Data Engineer **| Technologies:** Apache Spark, PySpark, Hadoop, Python

- Designed a scalable distributed-processing pipeline for large datasets using Apache Spark and PySpark.
- Performed complex transformations and analytics across the five V's of big data: volume, velocity, variety, veracity, and value.

Web Application Vulnerability Assessment | Cybersecurity

Role: Security Assessment **| Technologies:** Penetration testing, SQL injection, XSS, CSRF

- Conducted vulnerability assessments with industry-standard penetration-testing tools and validated SQL injection, XSS, and CSRF findings.
- Produced remediation reports with mitigation strategies and secure-coding recommendations.

Personal Portfolio | Full-Stack Web Application

Role: Full-Stack Developer **| Technologies:** MongoDB, Express.js, React.js, Node.js, CSS3 | Live site

- Developed a responsive portfolio to showcase engineering projects and technical writing.
- Deployed the application using standard DevOps practices for cross-device compatibility and fast load times.

Ankh-Arena | Fantasy Premier League Full-Stack Replica

Role: Full-Stack Developer **| Technologies:** React.js, Node.js, SQL, CSS | GitHub

- Built a responsive React interface and Node.js API endpoints with server-side application logic.
- Designed and managed SQL-backed data storage and responsive CSS styling.

LEADERSHIP AND ACTIVITIES

- Head of Technical Team, CDAFM, Alexandria Faculty of Medicine (2023-2024).
- Led the AAST team to 2nd place in the 2024 Intelligent Machines Championship at UCLan in the UK.
- Volunteer, Regional Forum on Education and Gender Inclusion in Energy Transition and Climate Action.

CERTIFICATIONS AND TRAINING

- The Complete Full-Stack Web Development Bootcamp - Udemy (In Progress)
- Sprints x Microsoft Summer Camp - Web Development
- Responsible AI with GitHub Copilot
- Introduction to Web Development with Blazor
- Implementing and Administering Cisco Solutions with Network Essentials
- Introduction to Programming Using Python - New Horizons
- Supervised Machine Learning Using Python - New Horizons